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Enhancing drone audition with rotor-conditioned deep models

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Abstract

The problem of recovering speech from audio recordings captured by a microphone aboard an unmanned aerial vehicle during flight is investigated. Enhancing a recording in this condition is difficult due to non-stationary noise from the motors and the propellers, along with environmental disturbance and motion-induced air flows. Together, these sources dramatically decrease the signal-to-noise ratio (SNR). This paper investigates the integration of rotor speed time series as a structured conditioning signal into neural speech enhancement models. We implement and evaluate rotor-informed variants of three state-of-the-art architectures: Wave-U-Net (time domain), DCCRN, and DCUNet (both time-frequency domain). Experiments on a custom UAV acoustics dataset spanning SNR levels from -30 to 0 dB show that rotor conditioning yields consistent and statistically significant improvements across SNR, SI-SDR, STOI, and PESQ metrics. These benefits generalize across model families, and a lightweight rotor-informed variant achieves best or near-best results despite using only 25% of the parameters. The findings establish rotor-informed conditioning as a robust and generalizable strategy for speech enhancement in low-SNR UAV environments.

1 Introduction

Unmanned aerial vehicles (UAV) or multirotor micro aerial vehicles (MAV), also known as drones, have recently become pervasive and are now used for mapping and localization in inaccessible areas. Research has focused mainly on developing high resolution camera-based solutions [1, 2]; however, UAVs with accurate acoustic sensing may play a critical role when vision is reduced by darkness, fog, smoke, or obstacles occluding the field of view [3]. Examples of UAV-based acoustic systems include drone audition platforms used for speaker localization [3], search-and-rescue [4], and environmental monitoring [5]. In these contexts, relatively little attention has been paid to the specific challenges of speech extraction and enhancement

involving autonomous systems that create noise due to, e.g., propellers and motors, the so-called *ego-noise* [6].

One main obstacle to effective acoustic sensing with UAV-mounted microphones is the loud ego-noise generated by drone motors and propellers [7]. Since these sources are closer to the drone microphones than a target human voice, speech quality is often significantly degraded. Along with the ego-noise, whose frequency profile depends on the rotation speed of each motor, natural and motion-induced air flows contribute to a further decrease in the SNR to unacceptable levels. Microphone array noise reduction methods thus become essential for enhancing the speech in a recording. However, at such speech signal levels, the performance of most array signal processing algorithms is considerably degraded [8]. At the same time, multi-channel techniques require microphone array hardware that is often cumbersome and too heavy as a load in compact drones [9]. Prior work has tackled ego-noise using signal processing techniques such as microphone array beamforming [10], multichannel Wiener filtering [11], and single-channel approaches based on implicit or GMM-based Wiener filters [12, 13].

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Ego-noise is always present in UAV-mounted microphone recordings. It depends on the number of rotors and their rotation speeds, which are constantly encoded within parameters available at runtime from the UAV control system. These parameters can be extracted from the commands that are sent to the electronic speed controllers, or conversely they can be decoded from the time series of the rotation speeds. Both can be used to feed a neural network during inference; however, runtime decoding reports more accurately than commands about the UAV's operating state and, hence, its ego-noise [14]. Rotor speed has been leveraged in other UAV contexts, such as fault detection [15], system modeling [16], and ego-noise synthesis [14], yet it has not been exploited as a conditioning signal in deep speech enhancement pipelines.

Most speech enhancement approaches have been formulated as time-frequency (TF) or spectrogram representations of mixture signals estimated from a waveform using the short-time Fourier transform (STFT) [17]. While STFT-based methods dominate the field, recent end-to-end models have challenged their supremacy in extremely low-SNR or non-stationary conditions, especially in real-time or streaming setups [18, 19]. In this work, we remain agnostic about the optimal representation domain. Instead, we focus on the effect of conditioning with rotor speed time series across both TF-based and waveform-based architectures, aiming to assess its contribution to speech enhancement independently of the domain.

To systematically evaluate the impact of rotor speed as an auxiliary signal, we implemented rotor-informed conditioning in three architectures: Wave-U-Net (WUN) [20], Deep Complex Convolution Recurrent Network (DCCRN) [21], and Deep complex U-Net (DCUNet) [22]. These models were chosen to represent a diverse range of input domains and processing strategies, and were recently benchmarked for UAV speech enhancement [9]. This architectural diversity enables us to test whether the rotor input consistently improves performance irrespective of model family and representation domain. Inspired by conditioning strategies in generative and causal audio models [23], we designed a rotor-processing pipeline and integrated it into each model. This architecture, adapted from prior work in UAV actuator fault detection [15], encodes the time series of revolutions per second (rps from now on) from individual UAV rotors, allowing the network to learn a structured representation of ego-noise patterns.

The motor data time series depends on the number of rotors. In most cases this series is manually synchronized with the UAV audio recordings [24]. Then, it is used to enrich the features extracted from the sequences

feeding the network at each level of the hierarchy. The resulting model does not require spectral analysis. The compactness of the model after the training and the short inference time bode well for experiments with processors onboard a UAV. Specifically, our network was trained and then tested using an ad-hoc dataset created by mixing different audio materials consisting of indoor sounds of clean speech, environmental acoustic disturbances, and the noise of a flying UAV (see Figs. 1 and 9), as well as its motor speed.

The main novelty of our work lies in incorporating a rotor-processing pipeline into a family of state-of-the-art single-channel speech enhancement architectures. This auxiliary module continuously processes rps data from UAV rotors, enabling each model to condition its audio inference on structured information reflective of ego-noise dynamics. We implemented and systematically evaluated this conditioning strategy across three diverse architectures: WUN (time domain), DCCRN, and DCUNet (both complex-valued time-frequency models). Our experiments show that rotor conditioning is not only broadly compatible but also generally preferable. Among 80 direct comparisons between rotor-informed models and their full-capacity baselines (except for a lightweight variant), the rotor-informed versions yield statistically significant improvements in 55 cases, confirming the consistent benefit of rotor input across domains and model families. Moreover, in 13 out of 16 metric-SNR configurations, the best-performing model includes rotor conditioning. Notably, even a compact rotor-informed model with just 25.3% of the parameters of the full DCCRN achieves the best performance in 3



Fig. 1 The MikroKopter quadrotor UAV used for the DREGON dataset, with the 3-D printed 8-channel microphone array replicated in the ALVAR room at the University of Udine in Pordenone, Italy. The picture refers to [24], and it is freely available in the additional material of the dataset at <http://dregon.inria.fr/datasets/dregon/>

configurations, and ranks among the statistically indistinguishable from best models in 2 more, underscoring the effectiveness of rotor integration even under strong architectural compression. These results affirm the practical utility of rotor signals in enhancing UAV audition across both time and frequency representations.

The paper is organized as follows: in Sect. 2, we briefly describe the relevant neural network-based speech enhancement methods, particularly those using motor speeds as additional information. In Sect. 3, we formalize the signal model and figure out the dependency of the ego-noise on the rotation speed. In Sect. 4, we describe the speech enhancement models. Section 5 shows results from the experimental tests. Section 6 discusses these results, highlighting practical implications, limitations, and potential avenues for future research. Conclusions follow in Sect. 7, also pointing to future work and potential improvements.

2 Related works

Early approaches to drone-based speech enhancement focused on classical signal processing techniques such as beamforming and multichannel Wiener filtering [11]. These methods rely on spatial diversity from microphone arrays and perform well in controlled conditions, however they are often impractical for compact UAV platforms due to weight, power, and form factor constraints [25]. More recently, neural networks have shown the ability to model temporal dependencies in audio signals [26] and have been adapted for speech enhancement tasks at very low SNR levels [8, 27]. In the context of UAVs, the application of deep learning is still relatively new, but recent benchmark studies have begun evaluating neural architectures under the challenging conditions of drone ego-noise and environmental interference [9].

Neural architectures for speech enhancement vary widely in how they represent and process the input signal. Most approaches operate in the TF domain, where the spectrogram magnitude is estimated and used to suppress noise [17]. While this offers interpretability and computational benefits, it neglects phase information that may carry useful cues in complex noise environments such as those found in UAV recordings [28]. Time-domain models such as WUN [29] and Conv-TasNet [30] address this issue by learning to map noisy waveforms directly to clean ones, potentially capturing fine temporal structures. In the context of UAV speech enhancement, recent comparative studies have shown that TF-domain models like DCUNet tend to outperform time-domain models like DCCRN or TasNet in terms of perceptual quality and intelligibility [9]. Based on this, we selected three representative models—WUN (time domain), DCCRN, and DCUNet (TF complex

domain)—to evaluate whether auxiliary rotor speed signals further enhance the performances independently of input representation and model family.

Several time-domain architectures have been proposed as alternatives to STFT-based enhancement, aiming to preserve fine-grained temporal and phase information. WaveNet [23] was an early autoregressive model for raw waveform generation, later adapted to speech enhancement [31]. SEGAN [32] introduced an encoder–decoder adversarial structure for waveform-based enhancement. Building on these ideas, WUN [29] adopted a multiscale convolutional encoder–decoder design, using downsampling and upsampling blocks with skip connections to process audio hierarchically. It has also been adapted for real-time settings [33]. In our work, WUN was included among the selected models to explore how rotor speed information interacts with architectures operating without explicit frequency decomposition.

Encoder–decoder architectures are widely used in speech enhancement for their ability to compress and reconstruct audio representations while preserving important features through skip connections. TasNet [34] exemplifies this structure in the time domain, learning a high-resolution latent representation from raw waveforms and applying a masking operation before decoding. While generally effective in speech separation problems, recent UAV-focused studies have shown that UNet-based models, particularly those operating in the TF complex domain, often outperform TasNet in both enhancement quality and computational efficiency [9]. For this reason, we prioritize WUN, DCCRN, and DCUNet in our evaluation, as they represent a strong encoding–decoding platform with complementary design principles across input domains.

In traditional drone audition, ego-noise suppression is often aided by auxiliary information. Several works have used rotor data—such as motor speeds or actuator states—to estimate the power spectral density (PSD) of ego-noise across different flight regimes [35, 36]. These approaches improve classical filtering techniques like multichannel Wiener filtering, however, they typically rely on handcrafted models and array geometries. More generally, this use of internal motor state to suppress self-generated acoustic noise aligns with principles of audio-motor integration explored in other robotic contexts [37]. Other methods have employed rotor speed profiles to guide multistage filtering pipelines using Gaussian Mixture Models (GMMs) [13]. Despite this body of work, the integration of structured rotor data into end-to-end neural networks for speech enhancement remains largely unexplored. To the best of our knowledge, our study is the first to systematically evaluate how rotor rotation

speed time series can condition both time-domain and TF neural models for drone-based speech enhancement.

Despite progress in both signal processing and deep learning approaches for UAV-based speech enhancement, current drone audition methods generally are not multimodal. While ego-noise is highly correlated with the drone's internal state, particularly rotor dynamics, this structured information has not yet been integrated into end-to-end neural models. Existing rotor-aware systems are limited to PSD estimation, sensor-guided filtering, or fault detection tasks, and do not address the challenges of extremely low-SNR speech recovery using neural networks. Moreover, the effectiveness of rotor conditioning across model types and input domains remains unexplored. Our work addresses this gap by introducing a generalizable rotor-informed pipeline and testing it on diverse architectures operating in both the time and TF domains, with and without rps input. We further assess whether the rotor series contributes meaningfully to model predictions through feature attribution analysis, offering new insight into how rotor signals can be exploited in neural UAV audition.

3 Signal framework

The problem of single-channel speech enhancement can be formulated in terms of estimating a clean signal $s(t) \in \mathbb{R}$ in the time domain given the noisy mixture $x(t) \in \mathbb{R}$, with

$$x(t) = a(t) * s(t) + e(t) + v(t), \quad (1)$$

where $a(t)$ is the impulse response from the source to the sensor, $*$ is the convolution operator, $e(t)$ is the ego-noise of the autonomous system, $v(t)$ is undesired additive noise. In the STFT domain:

$$X(k, f) = A(k, f)S(k, f) + E(k, f) + V(k, f), \quad (2)$$

where k is the signal frame number and f is the frequency bin.

Noise radiated by rotors is a mixture of tonal and broad-band noise. Tonal noise defines a sequence of spectral components multiple of a fundamental frequency, while broad-band noise extends over a wide frequency range. A rotor's tonal noise can be further split into loading and thickness noise and is characterized by sharp tonal noise at the blade-passing frequency and its harmonics. The spectrum of the ego-noise generated by propellers rotating at constant rps has predictable characteristics—see the power spectra in Fig. 2a, b, showing peak components respectively when one or four propellers rotate at 70 rps. In these conditions, Wiener filtering can be effective [38]. Broad-band noise is due to turbulence effects and non-linearities in the flow [39]. It originates from various sources including, among others, turbulent leading or trailing edge of the boundary layer and separation effects. Due to multi-rotor interactions and unsteady operations, broad-band noise in UAVs is generally unpredictable and can be as loud as tonal noise.

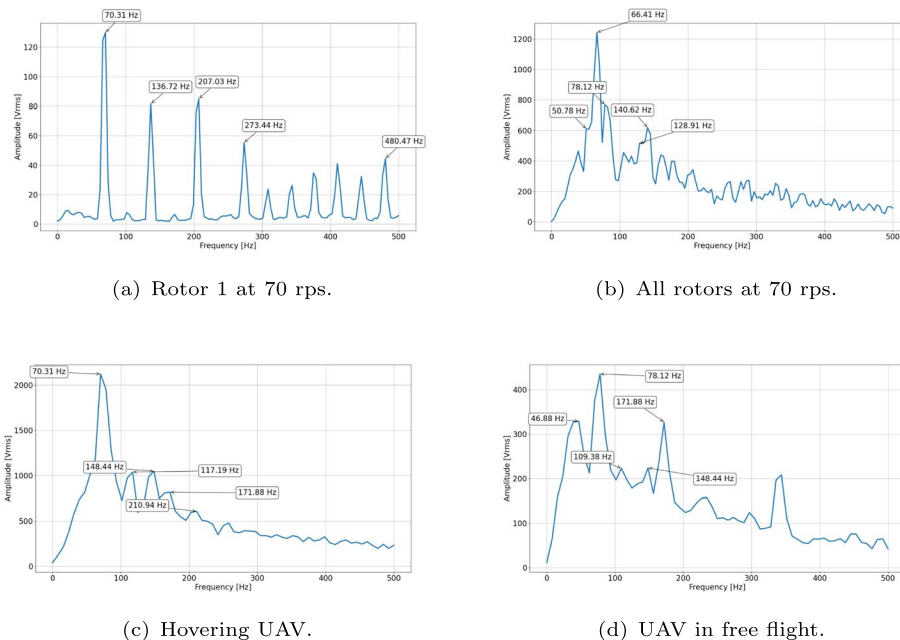


Fig. 2 Power spectrum of one second of ego-noise with rotors at different rps and in different UAV states

The overall noise spectral characteristics hence change their predictability depending on the UAV state—see Fig. 2c, d).

The spectrograms in Fig. 3 clarify the situation; tonal noise prevails only when a UAV is not in flight, with one propeller rotating at a constant speed. When all rotors rotate at the same speed and the UAV is not in flight, tonal components become less noticeable. Their temporal evolution during free flight becomes invisible from the spectrograms, reflecting the rotors' continuously fluctuating rps. These observations motivate the inclusion of rotor speed as an auxiliary input for neural models: while tonal components may be predictable from rps under static conditions, broadband ego-noise emerges from more complex rotor interactions and transitions, where rps can still serve as a latent descriptor of the UAV's dynamic state.

Formal analyses support the above empirical considerations. The Ffowcs Williams-Hawkins (FW-H) equation [40] is used in aeroacoustics to predict the sound radiation from an object traveling at subsonic speed. The Farassat formulation 1A [41] is typically used to find approximated solutions to the FW-H equation, by explicitly including among other parameters the blade geometry, the visco-elastic properties of the medium, and especially the blade surface angular velocity which corresponds to the rps information we aim at providing the network with. This formulation requires the use of numerical techniques since it does not provide a closed-form analytical solution.

Accurate simulations are frequently needed to estimate the tonal and broad-band noise of the UAVs' propellers and rotors for active noise control [42]. These simulations are computationally and memory intensive, in general, unsuitable for real-time implementation unless some reduction is applied at the cost of degrading the broadband noise modeling, even in a single rotating propeller scenario [43]. In practical terms, while these models describe the acoustic radiation from rotor dynamics

well, they cannot be computed in real-time by inference systems processed onboard a UAV. For this reason, in the following of this section, we will model the inclusion of the information about rps directly in (1).

We formulate the contribution of each rotor to the ego-noise $e(t)$ in terms of a tonal noise and a broad-band component: $e(t) = \sum_{r=1}^R e_r(t) + b(t)$, where R is the number of rotors, $e_r(t)$ is the tonal noise produced by the r -th rotor, and $b(t)$ is the broad-band noise produced by the interaction among rotors along with other nonlinearities that are inherent to the autonomous system movements. In the STFT domain:

$$E(k, f) = \sum_{r=1}^R E_r(k, f) + B(k, f), \quad (3)$$

Introducing the array of rps series $\omega(t) = (\omega_1(t), \dots, \omega_R(t))$, with $\omega_r(t)$ the rps of the r -th rotor, and assuming this array to be constant within each STFT time frame, we can explicit the ego-noise dependency from the rps in the transformed domain as

$$E(k, f, \omega(k)) = \sum_{r=1}^R E_r(k, f, \omega_r(k)) + B(k, f, \omega(k)), \quad (4)$$

Plugging (4) into (2), we have

$$X(k, f, \omega(k)) = A(k, f)S(k, f) + E(k, f, \omega(k)) + V(k, f), \quad (5)$$

and finally

$$S(k, f) = \frac{X(k, f, \omega(k)) - E(k, f, \omega(k)) - V(k, f)}{A(k, f)}. \quad (6)$$

This formulation aligns with the structure of TF neural models such as DCCRN and DCUNet, which operate on complex-valued STFT coefficients and can be

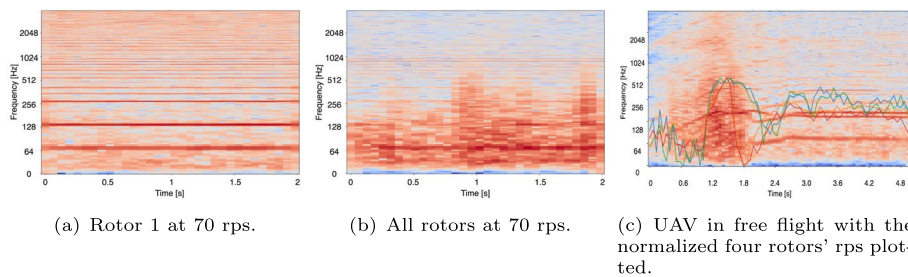


Fig. 3 Spectrograms of recordings taken from a UAV-mounted microphone with rotors at different rps and in several UAV states. Figures 3a and 3b are spectrograms of recordings made with the UAV on the ground and rotors rotating at constant speed. Figure 3c is the spectrogram of a recording made with the UAV in free flight, with the rps array superimposed

conditioned frame-wise with auxiliary information such as rps sequences.

Applying an inverse STFT to (6), we obtain the formulation of the target signal in the time domain, that we will use with our rotor-informed WUN model. In the following, we describe how rotor speed sequences can be used for conditioning both time-domain and time–frequency deep learning models for UAV speech enhancement.

Other methods have been developed to reduce non-stationary UAV ego-noise, including those based on Gaussian processes [44], dictionaries [14, 37, 45], and neural networks [46]. However, they have been tested at SNRs outside ranges applicable to a UAV, furthermore showing results that did not include objective measures of perceptual audio quality.

4 Rotor informed architectures for UAV speech enhancement

4.1 Overview of model families

We evaluate three families of speech enhancement architectures—WUN, DCCRN, and DCUNet—for their suitability in rotor-noise conditions encountered during UAV flight. These models are selected for their compactness, and ability to operate in either time or TF domains. To formalize the task, let $x[n] \in [-1, 1]$ be the noisy input signal, and let $\omega[m] \in \mathbb{R}^R$ be the corresponding rotor rps time series, sampled at a lower rate. The enhanced clean speech $\hat{s}[n]$ is obtained as a nonlinear transformation of both inputs:

$$\hat{s}[n] = f(x[n], \omega[m]), \quad (7)$$

where $f(\cdot)$ is implemented by one of the neural architectures described above.

Depending on the model family, f is learned in either the time domain or the TF domain. In the case of WUN, both $x[n]$ and the upsampled rps series $\tilde{\omega}[n]$ are directly processed, and the output $\hat{s}[n]$ is directly estimated in the time domain. Conversely, DCUNet and DCCRN operate in the complex TF domain. The noisy signal is first transformed via STFT:

$$X(k, f) = \text{STFT}\{x[n]\}, \quad (8)$$

and the network learns a complex spectral mask $M(k, f)$ or enhanced spectrum $S(k, f)$ using both $X(k, f)$ and $\tilde{\omega}[n]$. The estimated clean speech is then reconstructed via inverse STFT:

$$\hat{s}[n] \approx \text{STFT}_k^{-1}\{S(k, f)\}[n]. \quad (9)$$

In both cases, the goal is to learn an optimal mapping from noisy speech and rotor dynamics to clean speech,

either directly in the waveform space or in the spectral domain.

The first step consists of upsampling and aligning the rps series with the noisy audio. This corresponds to interpolate the values of the discrete-time array $\omega[m]$ so as to obtain a new rps array $\tilde{\omega}[n]$ fitting the rate of the audio signals:

$$\hat{s}[n] = f(x[n], \tilde{\omega}[n]). \quad (10)$$

In all rotor-informed architectures considered, both $x[n]$ and $\tilde{\omega}[n]$ are first processed via dedicated neural pathways, often starting with one-dimensional convolutions, before being fused at intermediate or bottleneck layers.

In all rotor-informed architectures considered, both $x[n]$ and $\tilde{\omega}[n]$ are processed via dedicated neural pathways before fusion. The specific fusion point and strategy vary by model family and are detailed in the following subsections.

4.2 Rotor integration strategies

As introduced in Sect. 3, the rps time series $\tilde{\omega}[n]$ acts as a proxy for the ego-noise structure. Its integration within the enhancement models is motivated by the physical model in Eq. (6), where the clean signal is computed by subtracting rotor-induced noise from the noisy observation. In neural architectures, this motivates the use of rps sequences as conditioning inputs, leading to learned mappings of the form:

$$S^*(\kappa, \varphi) = X^*(\kappa, \varphi, \Omega^{**}(\kappa, \varphi)), \quad (11)$$

where the clean speech representation S^* is estimated from the mixture and rotor representations via convolutional or recurrent blocks. These formulations justify the architectural decision to process and inject rotor features at specific temporal depths.

Each model family incorporates rotor information through a dedicated processing pipeline designed to extract relevant features from the rps time series $\tilde{\omega}[n]$. The processing depth and structure are adapted to the domain (time or TF), the model architecture, and the target fusion strategy.

Rotor speed sequences are integrated into the enhancement networks using three general fusion strategies, which differ in how and where the rotor features influence the audio representation:

- **Direct bottleneck projection:** rotor features are projected to match the audio bottleneck representation and added directly. This approach is efficient and exploits the global context without requiring recurrent modeling.

- **Recurrent fusion:** rotor and audio features are combined at the model bottleneck and jointly processed by a recurrent unit. This enables temporal alignment between rotor dynamics and audio content, supporting robust enhancement under fast-changing UAV noise.
- **Hierarchical injection:** rotor features are progressively injected at multiple depths within the encoder hierarchy. This allows the model to modulate feature extraction at different abstraction levels, depending on the rotor input.

Time-domain and time–frequency models employ these strategies differently. The following subsections describe their implementation in each architecture.

4.3 Rotor processing architectures

Rotor-processing architectures are based on the modeling assumptions in Sect. 3, where rotor dynamics are shown to influence ego-noise structure in both tonal and broadband components. Equation (6) formalizes this dependency in the STFT domain, while Eq. (11) provides a learned counterpart that maps noisy observations and rotor features to clean speech estimates. These dual perspectives motivate dedicated rotor pathways designed

to extract, align, and fuse relevant temporal features within each architecture.

Each model integrates rotor speed sequences through a dedicated processing pipeline. In time-domain architectures, the rps input is processed via CNN and recurrent layers, while in TF-domain models, fusion strategies vary across architectures and include direct projection, recurrent bottleneck fusion, or hierarchical injection. These approaches reflect trade-offs between temporal resolution, computational cost, and dynamic alignment needs. Details for each integration scheme are provided below and illustrated in Figs. 4, 5, 6, 7, and 8.

WUN with CNN-LSTM rotor processing (RPS-WUN). In the time-domain WUN architecture, rotor sequences are processed through a lightweight CNN stack with batch normalization and LeakyReLU activations. The output is passed through an LSTM layer and projected via a fully connected layer. The resulting features are injected hierarchically at multiple levels of the audio encoder via additive skip connections. This design allows the rotor dynamics to progressively influence feature extraction. The conditioning strategy mirrors that of WaveNet and is well suited to the long temporal context available in time-domain modeling (see Fig. 4).

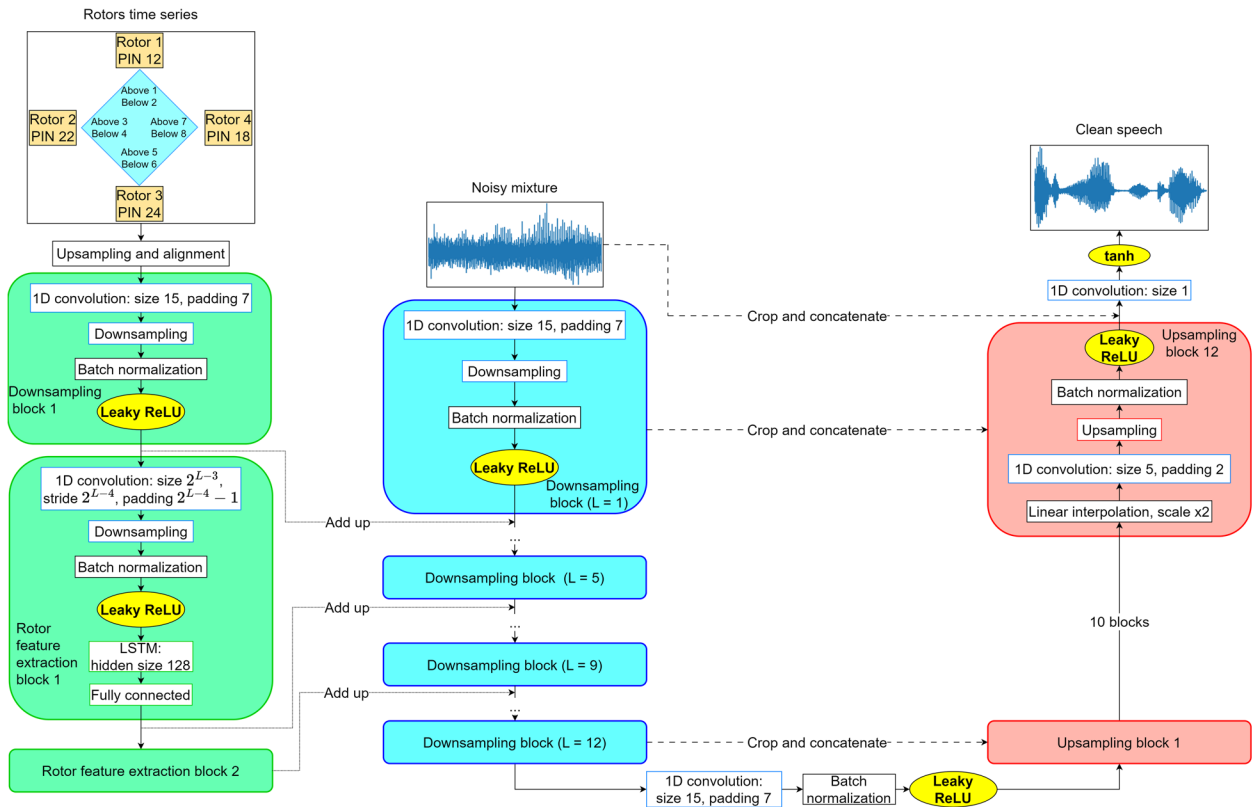


Fig. 4 Architecture of the rotor-conditioned WUN, using hierarchical CNN-LSTM rotor processing and time-domain convolution

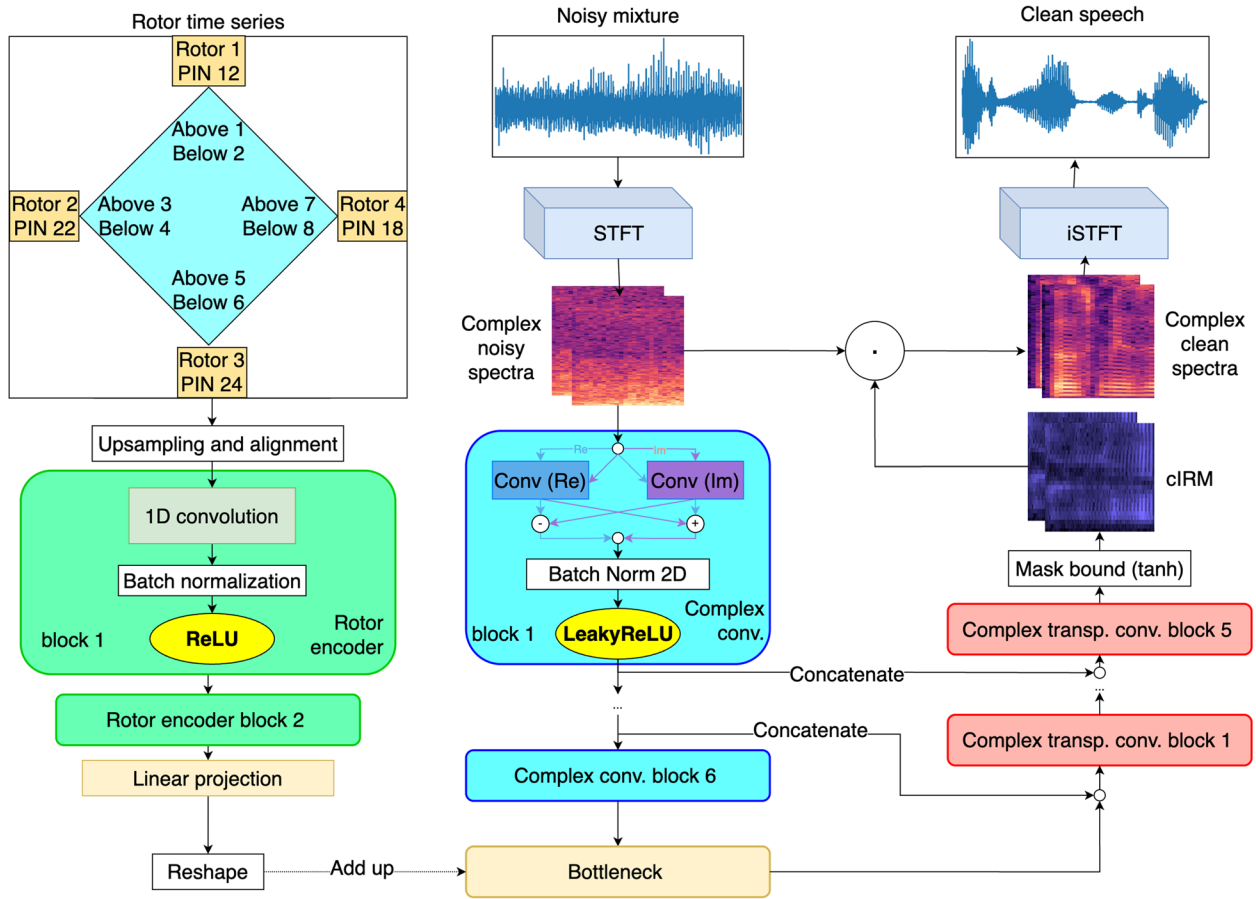


Fig. 5 DCUNet with projection of rotor features added at the bottleneck

DCUNet with bottleneck projection (RPS-DCUN6-P). In this DCUNet variant, we use a 6-layer encoder to obtain a compact audio representation. While this deep architecture provides increased receptive field and higher selectivity in frequency, it also aggressively downsamples temporal resolution—potentially limiting the alignment with high-frequency rotor dynamics. To integrate rotor information, the rps time series is processed by a convolutional network and projected via a linear layer to match the dimensions of the encoded audio bottleneck. The projected features are then additively fused with the audio representation. This approach—referred to as bottleneck projection—introduces minimal computational overhead and avoids recurrent modeling, making it a lightweight rotor conditioning strategy (see Fig. 5).

DCUNet with GRU-based fusion (RPS-DCUN5). This 5-layer variant of DCUNet preserves sufficient temporal granularity to enable effective recurrent modeling. Rotor features are extracted via a CNN encoder, then concatenated with the encoded audio bottleneck representation. The combined sequence is passed through a bidirectional GRU and projected back to the original

shape before decoding. This strategy enables the model to capture dependencies between rps patterns and audio over time, enabling dynamic modeling (see Fig. 6).

DCUNet with hierarchical rotor injection (RPS-DCUN5-H). In a third DCUNet variant, rotor features are progressively injected across encoder layers. At each of the five encoder levels, the rps input is processed through a convolutional block composed of a 1D convolution followed by a ReLU activation. The resulting features are interpolated to match the temporal resolution of the corresponding encoder layer and added to the encoder output before the next convolutional block, enabling distributed conditioning throughout the encoder hierarchy. This allows the network to learn which representation depth benefits most from rotor conditioning, and aligns with the idea that ego-noise contributions emerge at different abstraction levels (see Fig. 7).

DCCRN with GRU-based rotor fusion (RPS-DCCRN). In the rotor-informed DCCRN architecture, rotor sequences are processed via a CNN encoder and concatenated with the spectro-temporal audio features at the bottleneck. This fused representation is fed

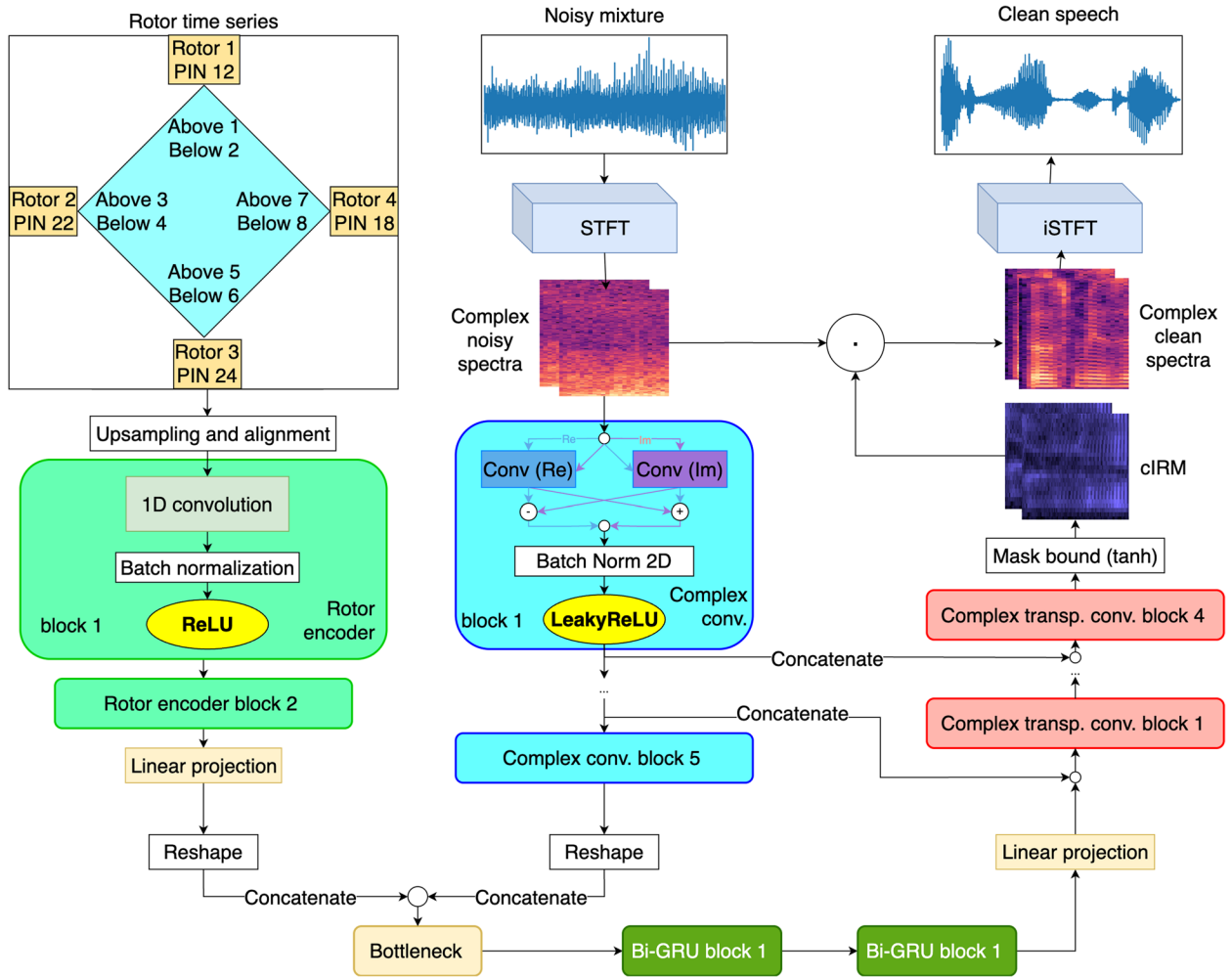


Fig. 6 DCUNet with GRU-based rotor–audio fusion at the bottleneck

into a GRU layer that models temporal dependencies across both modalities. The GRU output is projected and passed to the decoder. This design preserves the core architecture of DCCRN and leverages its recurrent module to adapt the enhancement based on rotor-induced variation over time (see Fig. 8).

Lightweight DCCRN for embedded deployment (RPS-DCCRN Lite). A reduced version of DCCRN is also explored for UAV-embedded inference. This model uses the same GRU-based rotor fusion strategy but with reduced channel counts and fewer parameters across all layers. This allows real-time inference on constrained hardware while preserving the benefits of temporal modeling and rps integration.

5 Experiments

We evaluated multiple speech enhancement models, including time-domain and time–frequency architectures, on a composite dataset whose core is the DREGON dataset [24]. It contains in-flight audio recordings made during different flying conditions (hovering, free flight, etc.) and the rps series of the four UAV rotors. Additional ego-noise material increasing the granularity of the rps series was recorded specifically for this study in the ALVAR room at the University of Udine in Pordenone, Italy. To keep the recording conditions of the additional material as similar as possible, we replicated the DREGON cubic array and then took multi-channel recordings with it. Such recordings are also left for future development of multi-channel noise reduction models applicable to drone listening.

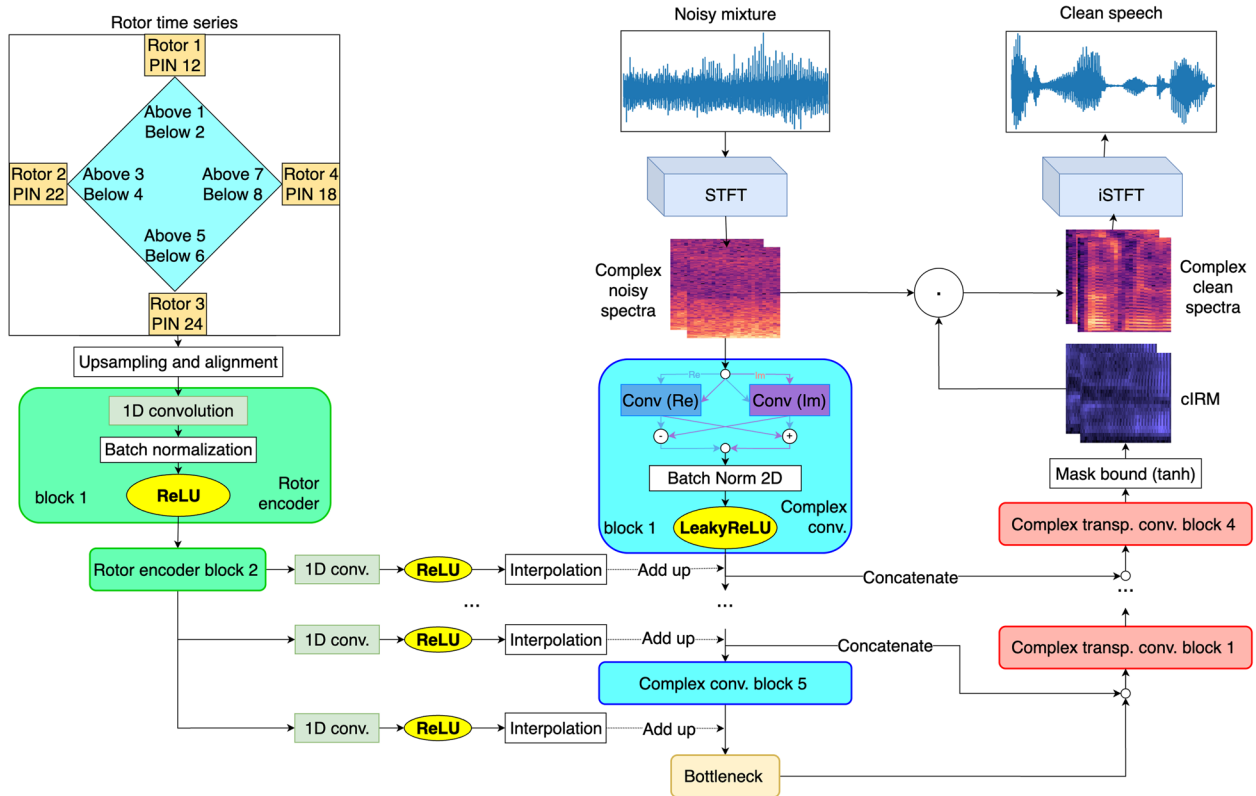


Fig. 7 DCUNet with hierarchical rotor feature injection across encoder levels

5.1 Dataset

One target speech, one instance of environmental noise, and two different instances of quadcopter ego-noise formed respectively the training, validation, and testing datasets. The target speech was extracted by the TIMIT corpus [47]. The environmental noise, taken from the Diverse Environments Multichannel Acoustic Noise Database (DEMAND) [47], was recorded in nine different environmental conditions using a 16-channel microphone array. The reverberation time in the DREGON recordings is under 150 ms, which is consistent with near-field open-space UAV conditions. The contribution in terms of additional noise of this reverberation is negligible when mixed with DEMAND environmental noise, thus justifying the use of synthetic mixtures combining these sources.

The ego-noise from the DREGON dataset contains clean speech, speech-free noise, and noisy in-flight audio recordings made with a cubic 8-microphone array at 44.1 kHz. The additional ego-noise recorded at the ALVAR room consisted of takes of the four rotors rotating individually and then altogether at a 5-s stepwise increasing speed, covering the entire operating range from the lowest value of 36 rps to a maximum of 84 rps. These takes were recorded by cloning the DREGON microphone

array and mounting the clone on a quadrotor as in Fig. 9. The clone was provided with eight MEMS omnidirectional microphones and then mounted on a custom quadcopter based on a DJI F450 frame, a Pixhawk PX4 Autopilot flight controller, and a Raspberry Pi 3 companion single-board computer (SBC). The companion SBC runs software allowing independent rps control of each rotor and simultaneous recording of the rotation data. Rounds were counted thanks to an optical device sensing color transitions on the rotating motor, which was alternatively colored black and white. The resulting rps time series were manually synchronized with the audio recordings. Eight-channel recordings at 32-bit resolution and 48 kHz sampling rate were taken using a MiniDSP USBStreamer I2S-to-USB audio interface, hosted on a computer running Audacity.

The various instances of ego- and environmental noise were downsampled at 8 kHz, then synthetically mixed at several intensity levels and in combinations with different percentages depending on the specific dataset partition.

The resulting training dataset contains audio recordings in the following proportions: clean speech 1%, ego-noise 80%, environmental noise 0.1%, and single-rotor ego-noise 5%. The rest of the recordings are noisy mixtures as described later. Clean speech consists of 4158 utterances

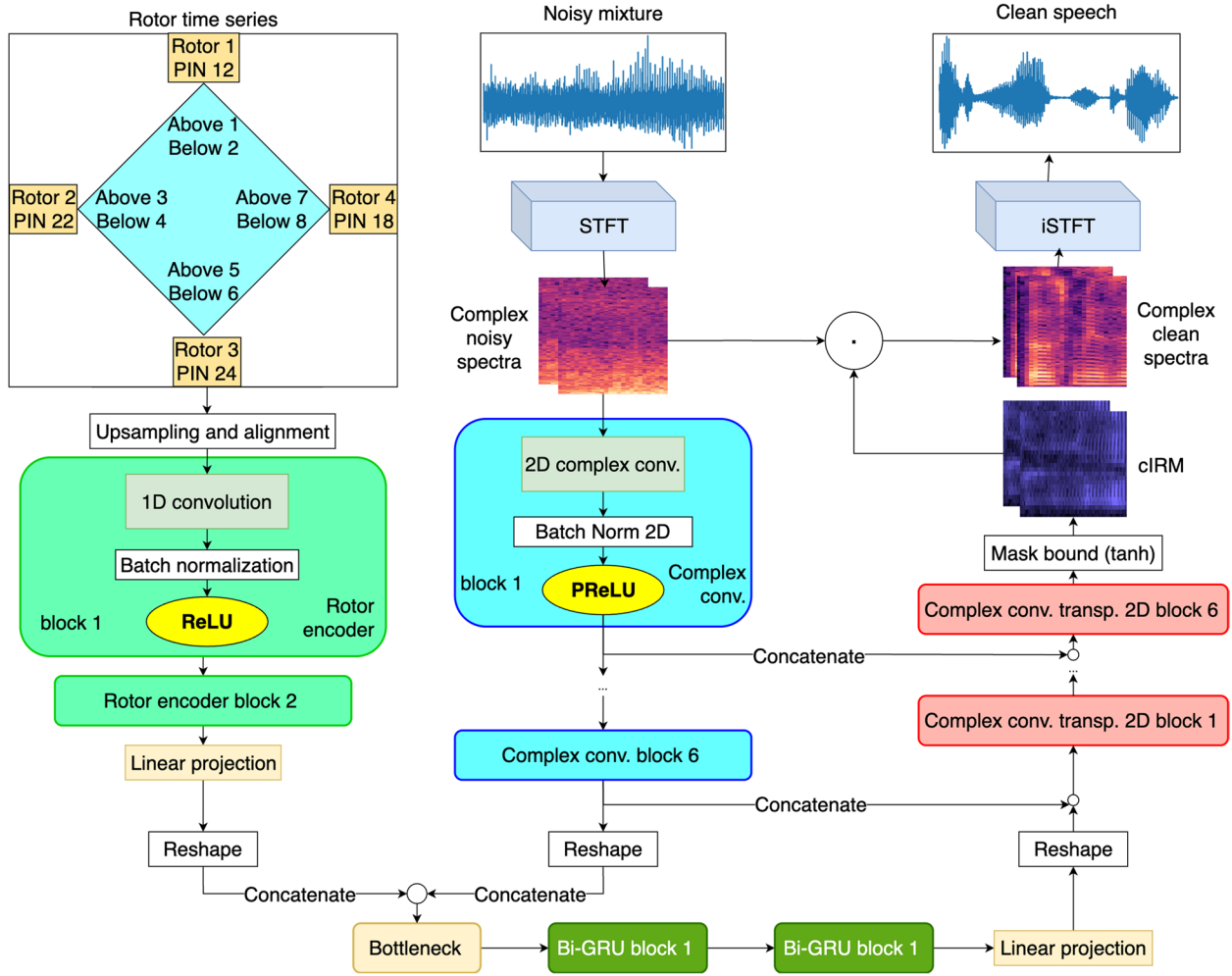


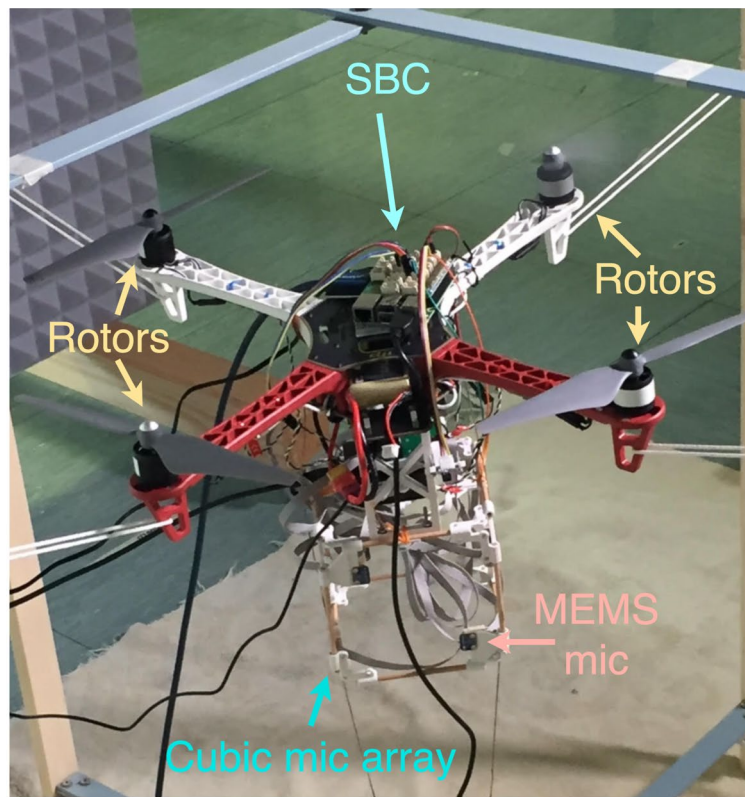
Fig. 8 Rotor-informed DCCRN architecture with GRU-based fusion at the bottleneck

(i.e., 90% of the TIMIT training subset), in which a segment of 16,384 samples was arbitrarily cropped for every utterance. Ego-noise is made with three DREGON recordings (namely, *DREGON_rectangle_nosource_room2*, *DREGON_spinning_nosource_room2*, and *DREGON_updown_nosource_room2*) along with the DREGON individual rotor recordings. Two noise segments having the same length were randomly taken from the ego-noise recordings and three DEMAND environmental noises, then they were mixed by selecting SNR levels respectively equal to 20, 15, 10, and 5 dB. The two audio sequences were mixed by sampling SNR uniformly across the range $[-25, -5]$ dB, in which the SNR of the noisy sequence is defined as

$$\text{SNR} = 10 \log_{10} \frac{\sum_{n=1}^T s^2[n]}{\sum_{n=1}^T v^2[n]}. \quad (12)$$

The validation dataset uses the remaining 10% of the TIMIT train subset with the same percentages of ego-noise, environmental noise, and single-rotor ego-noise recordings. Here, the ego-noise was taken from the DREGON tracks recorded in room 1 and the DREGON individual rotor recordings, conversely the environmental noise was taken from the three DEMAND recordings left. The target speech and the noise were added by uniformly selecting SNR levels respectively equal to -25 , -15 , -10 , and -5 dB.

The clean speech in the test dataset consists of all the 1680 utterances of the TIMIT test subset. The noise was drawn from two unseen ego-noise recordings of the DREGON dataset (namely, *DREGON_free-flight_nosource_room2* and *DREGON_hovering_nosource_room2*), the supplementary ego-noise recordings, and the three unseen DEMAND recordings left. Each utterance was mixed with a randomly cropped noise segment at four SNR levels equal to -30 , -20 , -10 , and 0 dB. In all



(a) The custom quadcopter with the SBC and the cubic 8-microphone array replica.



(b) Detail of the white stripe on the black-colored rotating motor for the optical-based rps detection.

Fig. 9 Setup at the ALVAR room

datasets, the default crop length approximately amounts to 2 s corresponding to 16,384 samples at 8 kHz.

5.2 Experimental setup

All deep learning models—WUN, DCCRN, DCUNet, and their rps-informed variants—were trained on the same dataset by setting a batch size to 8, and early stopping based on validation loss stagnation for ten consecutive evaluations. The WUN models were optimized using Adam with an initial learning rate equal to 0.001, reduced by a factor of 0.1 after 15 epochs without improvement. DCCRN and DCUNet used the Adam optimizer with an initial learning rate of 0.0003 and the same reduction and early stopping

schedule. Training was performed for up to 300 epochs, with validation every 5 epochs and checkpoint saving every 10 epochs.

The GMM-based multistage Wiener filtering baseline was trained using the Expectation-Maximization algorithm from scikit-learn with a maximum of 100 iterations, employing 6 Gaussians for speech and 9 for noise, both with diagonal covariances. Input features were log-magnitude STFTs normalized in the power spectral domain. Enhancement used a two-stage iterative Wiener filtering procedure with gain parameters $\beta = 2.0$ and $\gamma = 1.0$. STFTs were computed using a Hann window (512-point FFT, 256-sample hop, 8 kHz sampling rate).

We also included TasNet as a lightweight, time-domain baseline. It was trained for 300 epochs using Adam with an initial learning rate of 0.001, halved when the loss stopped improving across three epochs. Early stopping was triggered after ten stagnant validation loss evaluations. This baseline was retained for its runtime efficiency and simplicity, despite underperforming on UAV data.

The whole training was performed on a NVIDIA GeForce RTX 3090 graphic card inside a workstation in the Artificial Intelligence for Human-Robot Collaboration (AI4HRC) Lab at the University of Udine.

5.3 Evaluation

The speech enhancement performance was assessed using objective metrics. Every enhanced speech signal was compared against the clean speech reference. The metrics are the SNR computed as in (12), the PESQ as proposed in [48], the STOI as implemented in [49], and the SI-SDR [50]. These metrics were considered as a comparative reference once applied to the signal when it was filtered using the Wiener method, as implemented in the PySound Python library¹. In addition to these measures, in Table 3 the inference time and model size are considered in terms of the number of the trainable parameters. Inference was performed on a 2.60-GHz Intel laptop operated by Windows. The source code and some audio examples can be downloaded at <https://github.com/gullogullo/Rotors-informed-Wave-U-Net-for-Speech-Enhancement>.

We evaluate all speech enhancement models on a test set composed of UAV-corrupted audio mixtures at four SNR levels: -30 , -20 , -10 , and 0 dB. Performance is measured using four standard metrics: SNR, SI-SDR, STOI, and PESQ. A Shapiro-Wilk test was performed to assess normality of the data distribution [51], furthermore a Levene's test was used to check homoscedasticity [52]. If normality and homoscedasticity hold, means of the measure variations are reported in the tables and the significance of their differences was tested with a one-way ANOVA [53]. Otherwise the medians are reported, and a Kruskal-Wallis H-test was used [54]; if the result is significant, a Dunn's post-hoc pairwise test for multiple comparisons of mean rank sums [55] was performed to find significant differences between the medians. The statistical analysis was performed using the SciPy Python package [56]. The best results are reported in bold when they are significantly different.

5.4 Architecture implementation details

All models were implemented and trained using PyTorch, with rotor-informed variants built on top of baseline structures. RPS-WUN is based on a 5-level encoder-decoder topology with symmetric 1D convolutions, transposed convolutions, and skip connections at each level. It accepts waveform inputs and is trained using MSE loss. The rotor pathway consists of two 1D convolutional layers (kernel size 3, 64 channels), batch normalization, LeakyReLU activations, one LSTM layer with 128 units, and a linear projection. Rotor features are injected additively into the encoder outputs every two layers. Optuna [57] was used to tune architectural hyperparameters. We compared 10 realizations of this optimization process by averaging the mean STOI and PESQ variations between the clean and the enhanced target speech of every validation item.

RPS-DCCRN operates in the complex TF domain. Inputs are transformed using a 512-point STFT with 256-sample hop, Hann window, and 8 kHz sampling. The encoder consists of six complex 2D convolutional layers with increasing channels from 32 to 512, followed by a two-layer bidirectional GRU with 256 hidden units. The decoder mirrors the encoder using transposed convolutions with skip connections. Rotor features are extracted via a 2-layer 1D convolutional block (kernel size 3, 64 channels, ReLU), interpolated to match the STFT length, and concatenated with the encoded spectrogram before the GRU. The model is trained with SI-SDR loss on the waveform reconstructed by inverse STFT.

RPS-DCCRN Lite adopts the same rotor fusion strategy as RPS-DCCRN but reduces the encoder-decoder to four convolutional layers with channels ranging from 16 to 128. It uses a single-layer bidirectional GRU with 128 hidden units and keeps the same rotor encoder. This configuration is optimized for real-time inference on embedded UAV platforms.

DCUN6 and RPS-DCUN6-P use a 6-layer U-Net architecture with 2D convolutions, doubling channels at each layer starting from 32, batch normalization, and LeakyReLU activations. The decoder mirrors the encoder with skip connections and instance normalization. RPS-DCUN6-P uses a rotor encoder composed of two 1D convolutional layers (kernel size 3, 64 channels) and a linear projection, which is added to the bottleneck representation. Training is performed with SI-SDR loss on the reconstructed waveform.

RPS-DCUN5 reduces the number of encoder-decoder layers to five. The rotor encoder matches that of RPS-DCUN6-P. Rotor features are concatenated with the bottleneck and passed through a bidirectional GRU with 256 hidden units, followed by a projection before

¹ <https://github.com/martinmcbride/pysound>

decoding. This variant improves alignment between rotor dynamics and STFT frames.

DCUN5 and RPS-DCUN5-H use the same 5-layer U-Net architecture with instance normalization. In RPS-DCUN5-H, rotor features are injected at every encoder levels using interpolated 1D convolutional outputs (kernel size 3, 64 channels), followed by a ReLU activation, and aligned with the temporal resolution of each encoder block.

TasNet was retained as a lightweight non-rotor baseline using its default structure. It uses an encoder with 500 filters of length 40, a separation module with 3 temporal convolutional blocks (each including dilated 1D conv, PReLU, and layer normalization), and a decoder with transposed convolution matching the encoder basis. Each block has 128 channels and uses residual connections. No rotor input is used.

5.5 Results

In Table 1, the median objective measure variations for the best-performing models across different SNR conditions and metrics are displayed; Table 2 presents the median percentage variations of rotor-informed models relative to their non-RPS baselines; and Table 3 reports mean inference times, averaged across test set enhancements, along with model sizes for all evaluated models. For illustration, we plot spectrograms of a sample utterance in Fig. 10, showing the noisy input, the clean reference, and the RPS-DCCRN Lite enhanced output.

Across all SNR conditions, the Shapiro–Wilk test confirmed consistent violations of normality. As a result, we adopted non-parametric statistical testing, using the Kruskal–Wallis H -test for group comparisons of medians, followed by Bonferroni-corrected Dunn’s post-hoc tests for pairwise significance ($\alpha = 0.05$). Across most conditions, rotor-informed models consistently ranked among the best performers. DCCRN-based architectures dominated overall, with both full and Lite variants achieving top performance across multiple metrics and SNR levels. RPS-DCCRN frequently held the top position or performed on par with the best, particularly in STOI and SNR improvement. The DCUNet family also exhibited strong performance even without rotor input, suggesting a particular robustness of that architecture. Nonetheless, rotor integration substantially improved DCUNet performance, especially at low SNR levels. WUN showed the largest relative improvements when augmented with rotor information, but its overall absolute performance scored below TF-domain models.

SNR improvement (see Fig. 11): At -30 dB, DCCRN achieved the best median SNR gain ($+29.9$), with RPS-DCCRN consistently ranking top across all levels. While RPS-DCCRN Lite scored lower in SNR

Table 1 Median of the objective measures variations of the best performing model per metric and SNR level. Models statistically superior to all others are reported in bold. In cases where the top model is not statistically different from others, all comparable models (with $p \geq 0.05$) are listed

SNR	Metric	Best model [Median]
SNR = -30 dB	SNR	DCCRN [29.931]
	SI-SDR	RPS-DCUN6-P [8.199] RPS-DCCRN [7.942]
	STOI	RPS-DCUN5 [−0.037]
SNR = -20 dB	PESQ	DCUN6 [0.122] RPS-DCUN6-P [0.095]
	SNR	RPS-DCCRN [21.063]
	SI-SDR	RPS-DCCRN [14.892] RPS-DCCRN Lite [14.201]
SNR = -10 dB	STOI	RPS-DCCRN [0.026]
	PESQ	RPS-DCCRN [0.170] DCUN6 [0.149]
SNR = 0 dB	SNR	RPS-DCCRN [15.180]
	SI-SDR	RPS-DCCRN Lite [14.029] RPS-DCCRN [13.823] DCCRN [13.705]
	STOI	DCCRN [0.074] RPS-DCCRN [0.062] RPS-DCCRN [0.296] DCCRN [0.285]
SNR = 0 dB	PESQ	RPS-DCCRN [11.422]
	SI-SDR	RPS-DCCRN Lite [11.613] RPS-DCCRN [11.368] RPS-DCCRN Lite [0.067] DCCRN [0.061] RPS-DCCRN [0.058] DCCRN [0.538] RPS-DCCRN Lite [0.529] RPS-DCCRN [0.513]

at 0 dB, it still performed well in SI-SDR and STOI. DCUNets (e.g., RPS-DCUN5/5-H) achieved statistically significant SNR improvements at low SNRs. RPS-WUN showed the largest relative SNR gain ($+99.1\%$ at -30 dB), though its absolute performance trailed TF-domain models.

SI-SDR (see Fig. 12): RPS-DCCRN consistently ranked among the top across all SNRs, reaching 14.9 dB at -20 dB and 13.8 dB at -10 dB. DCUNet models also benefited notably from rotor integration: RPS-DCUN6-P outperformed its baseline by $+57.8\%$ at -30 dB, while RPS-DCUN5 and RPS-DCUN5-H showed $+607.1\%$ and $+544.7\%$ relative improvements, respectively. RPS-WUN achieved a $+139.4\%$ gain at -30 dB.

Table 2 Median percentage variation of rps-informed models with respect to their non-rps baselines across SNR levels and metrics. Statistically significant improvements over the baseline ($p < 0.05$) are reported in bold

		RPS-DCCRN	RPS-DCCRN Lite	RPS-DCUN 6-P	RPS-DCUN 5	RPS-DCUN 5-H	RPS-WUN
SNR	SNR	−0.2	−10.3	3.0	6.7	6.2	99.1
=	SI-SDR	39.9	25.5	57.8	607.1	544.7	139.4
−30	STOI	44.4	3.3	−11.2	39.8	−30.4	−56.5
dB	PESQ	73.4	30.8	−21.9	41.2	−27.1	464.9
SNR	SNR	2.2	−19.0	0.5	0.4	1.2	55.4
=	SI-SDR	8.4	3.4	11.5	33.2	81.1	198.8
−20	STOI	629.3	30.7	−12.2	63.8	7.0	16.5
dB	PESQ	65.4	16.2	−32.9	4.6	−11.8	1422.7
SNR	SNR	12.0	−70.4	0.3	1.3	3.1	29.3
=	SI-SDR	0.9	2.4	7.2	0.7	1.1	302.1
−10	STOI	−17.0	−24.9	308.3	77.3	79.1	135.9
dB	PESQ	3.6	−13.6	7.2	13.5	22.0	737.3
SNR	SNR	77.7	−209.2	−5.1	2.4	−1.3	44.3
=	SI-SDR	5.0	7.2	6.4	−2.4	1.3	68.2
0	STOI	−4.9	9.6	81.4	−8.4	14.1	131.8
dB	PESQ	−4.5	−1.6	−3.5	−3.8	11.0	2194.3

Table 3 Inference time (in seconds) and model size (millions of parameters) for all evaluated models. Percentage variations (%) are computed for rps-informed models relative to their non-rps counterparts. The lowest inference time and model size are reported in bold

Model	Time [s]	Model size [M]	% Δ time	% Δ size
RPS-WUN	0.1131	15.4768	+97.0	+52.7
WUN	0.0574	10.1328	−	−
RPS-DCUN6-P	0.0068	8.1068	+113.8	+1.1
DCUN6	0.0032	8.0181	−	−
RPS-DCCRN	0.0210	25.1756	+5.4	+0.5
DCCRN	0.0199	25.0546	−	−
RPS-DCUN5-H	0.0043	4.8831	+58.6	+0.3
DCUN5	0.0027	4.8708	−	−
RPS-DCCRN Lite	0.0071	6.3416	+4.3	+1.2
TasNet	0.0842	8.5780	−	−
Wiener	0.0043	−	−	−
GMM	0.0036	−	−	−

STOI (see Fig. 13): At −30 dB, RPS-DCUN5 led all models in STOI, achieving the only statistically significant improvement (+39.8%), followed by RPS-DCCRN (+44.4%) and RPS-DCCRN Lite (+3.3%). At −10 dB, RPS-DCUN6-P achieved a remarkable +308.3% gain over its baseline. While RPS-DCCRN's absolute scores

remained high across SNRs, the relative STOI gains were most prominent in the DCUNet family.

PESQ (see Fig. 14): DCCRN and RPS-DCCRN achieved the highest PESQ scores overall, with DCCRN reaching 0.538 at 0 dB. At lower SNRs, rotor-informed DCUNets performed strongly: RPS-DCUN5-H achieved +22.0% gain at −10 dB, and RPS-DCUN5 had +41.2% at −30 dB. RPS-WUN showed a dramatic +1422.7% gain at −20 dB and +2194.3% at 0 dB, reflecting a very low baseline.

Efficiency (see Table 3): Rotor-conditioned models incurred modest overheads in inference time and model size. Most remained well below the 150 ms real-time threshold [58], except DCCRN. RPS-WUN nearly doubled inference time (+97%) and increased size by 52.7%, yet stood within acceptable limits. RPS-DCUN6-P had the largest relative inference time increase (+113.8%), but its absolute time (6.8 ms) was negligible. Aside from RPS-WUN, all other rotor-informed models showed size increases below 1.2%.

6 Discussion

The experimental results confirm that incorporating rotor rps information significantly improves speech enhancement algorithms. Rotor-informed models consistently outperformed their counterparts free of rps information, demonstrating broad compatibility and substantial gains across diverse architectures, metrics,

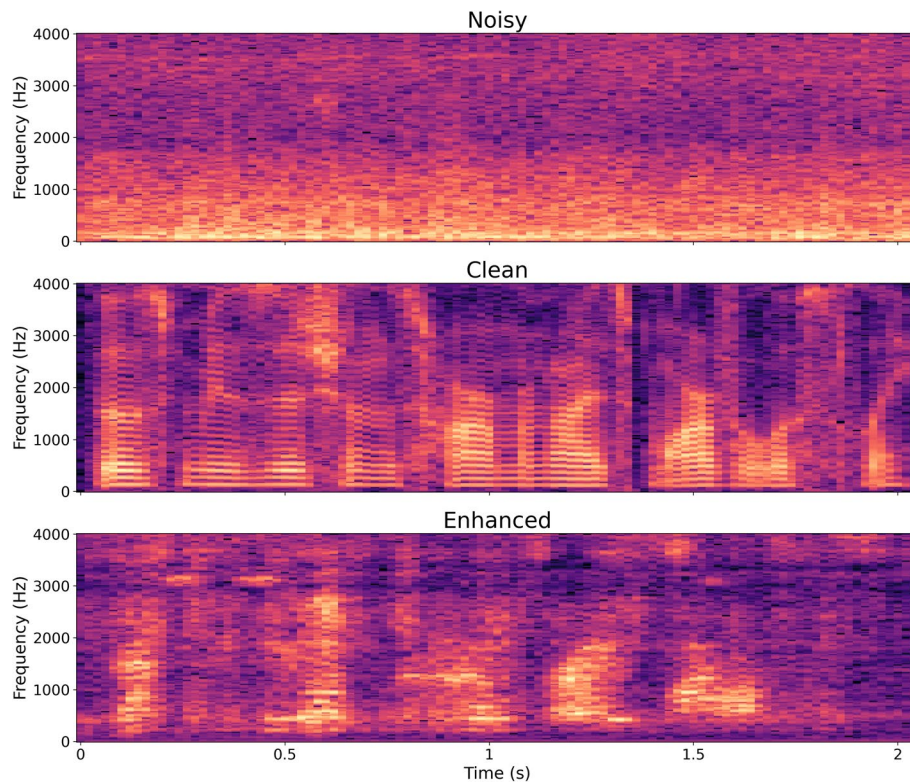


Fig. 10 Spectrograms of a noisy input mixed with SNR equal to -20 dB (top), the clean utterance (middle), and the enhanced RPS-DCCRNLite output (bottom)

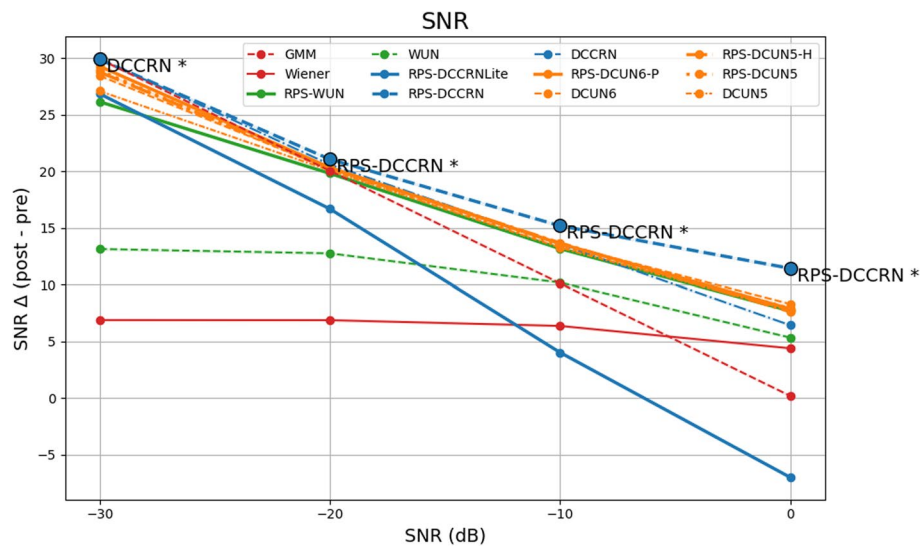


Fig. 11 SNR improvements (Δ post-pre) across SNR levels for all models

and signal-to-noise ratio (SNR) conditions. Specifically, the rotor-informed models yielded statistically significant improvements in 55 out of 80 direct comparisons against baseline variants, underscoring the robustness and practical utility of rotor conditioning.

Among the evaluated architectures, RPS-DCCRNN emerged as particularly effective. This approach, which integrates rotor features through CNN encoding, concatenation at the spectro-temporal bottleneck, and subsequent temporal modeling via GRU, consistently

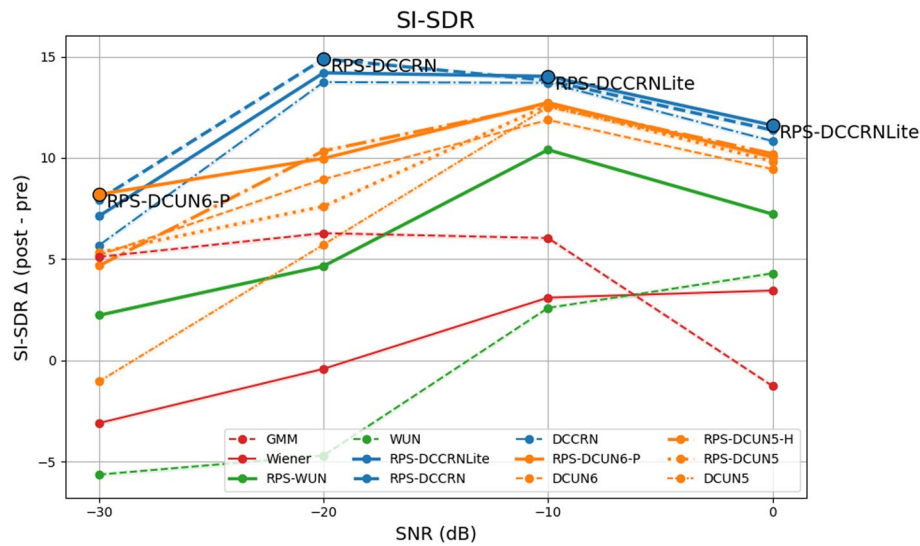


Fig. 12 SI-SDR improvements (Δ post-pre) across SNR levels for all models

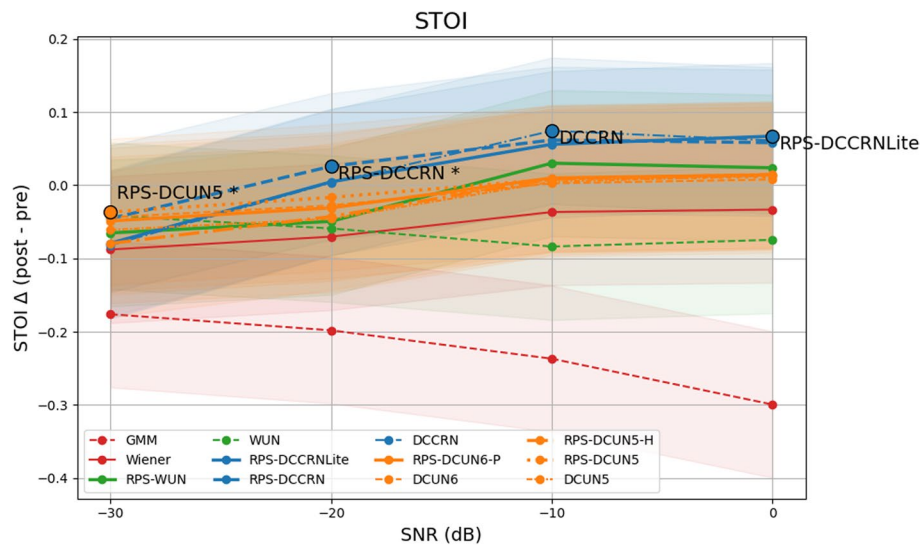


Fig. 13 STOI improvements (Δ post-pre) across SNR levels for all models

ranked at the top across nearly all SNR levels and metrics. RPS-DCCRN achieved the best SNR gains from -20 dB upwards (e.g., 21.063 dB at -20 dB, 15.180 dB at -10 dB, and 11.422 dB at 0 dB), alongside strong performance in SI-SDR and STOI measures.

Moreover, the lightweight variant (RPS-DCCRN Lite) demonstrated notable effectiveness despite significant architectural compression, using fewer parameters and achieving comparable or superior performance relative to more computationally demanding architectures. This compact model achieved best or statistically equivalent results in SI-SDR at several SNR conditions (e.g., 14.201 dB at -20 dB and 11.613 dB at 0 dB), suggesting that

rotor-informed architectures can be effectively downsized without substantial performance degradation—an essential feature for onboard drone applications. Such findings align with recent investigations that emphasize the critical importance of lightweight neural network designs for speech enhancement under drone noise, highlighting the feasibility of efficient models for constrained UAV environments [59].

The DCUNet architecture also benefited substantially from rotor integration, particularly at lower SNR conditions. Variants of DCUNet using different rotor-fusion strategies, such as bottleneck projection (RPS-DCUN6-P), hierarchical injection (RPS-DCUN5-H), and

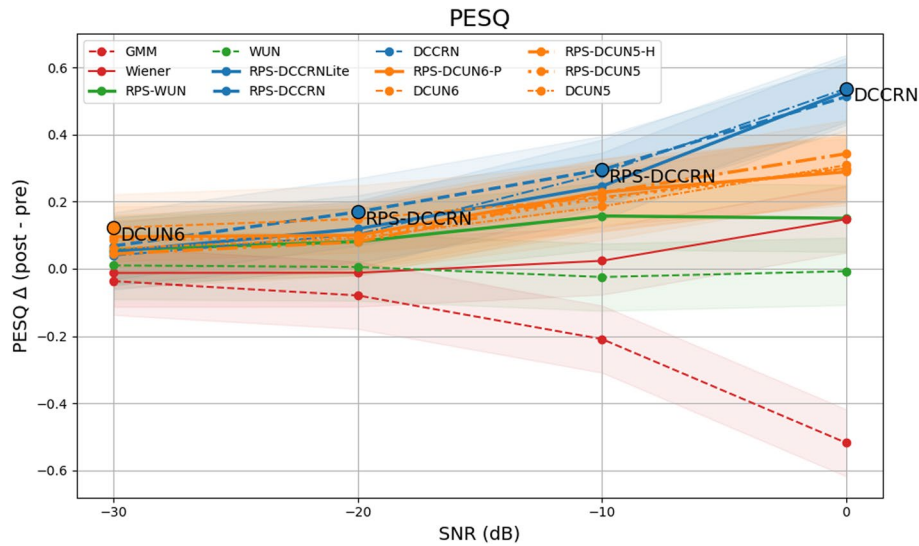


Fig. 14 PESQ score improvements (Δ post-pre) across SNR levels for all models

GRU-based fusion (RPS-DCUN5), each demonstrated scenario-specific advantages. For example, RPS-DCUN6-P provided a significant SI-SDR improvement (+57.8%) at -30 dB, and RPS-DCUN5 achieved the only statistically significant STOI enhancement at -30 dB (+39.8%). This variability indicates that optimal rotor conditioning may depend on the interplay between temporal granularity and model depth—highlighting the value of adaptive rotor-fusion methods across TF-domain models.

In contrast, RPS-WUN exhibited the largest relative improvements due to rotor conditioning (e.g., +99.1% in SNR at -30 dB and remarkable gains in PESQ scores up to +2194.3%). However, in absolute terms, it consistently trailed the TF-domain architectures (DCCRn, DCUNet), limiting its practical applicability. This result echoes previous findings that highlight the intrinsic advantages of TF-domain representations for UAV speech enhancement [9].

Considering real-world drone audition constraints, latency and computational complexity emerge as critical factors. Drone audition applications require real-time onboard processing, restricting model size, computational load, and future context length. Real-time speech enhancement frameworks typically impose stringent latency constraints, allowing algorithms unlimited access to past frames but strictly limiting future context to within 40 ms [60]. However, the evaluated models—both DCCRn and DCUNet architectures—process entire audio samples at once and are inherently non-causal, thus currently unsuitable for real-time deployment without

substantial modification. Future work must therefore explicitly adapt these architectures to strictly causal inference settings, ensuring compatibility with practical real-time drone audition requirements.

A significant limitation of the current study is the absence of explicit computational complexity metrics, such as FLOPs, which are crucial indicators for onboard deployment. Recent benchmarks demonstrate that models such as DCUNet impose substantial computational loads (112.136 GFLOPs), significantly exceeding lightweight alternatives optimized for constrained environments (11.617 GFLOPs) [59]. Without a precise FLOPs quantification, the deployment of the evaluated models in practice remains uncertain. Future studies should include comprehensive computational complexity analyses to conclusively determine model suitability for real-time UAV applications.

Another limitation is the exclusive use of synthetic datasets, which raises questions about generalization to real-world drone-recorded acoustic scenarios. Recent research has stressed that algorithms validated solely on synthetic mixtures may not directly translate to practical drone applications without adjustments based on real-world acoustic data [61]. Thus, future validations on real-world drone acoustic recordings are necessary to establish robust applicability.

In summary, while our results robustly support the integration of rotor rps information across multiple state-of-the-art architectures, addressing deployment-specific constraints (latency, complexity, FLOPs) and validating against real-world datasets

represent important next steps toward practical UAV deployment.

7 Conclusion

In this study, we investigated the integration of rotor speed signals into end-to-end neural speech enhancement architectures for drone audition. By conditioning three state-of-the-art models (Wave-U-Net, DCCRN, and DCUNet) with rotor time series, we achieved consistent and statistically significant improvements across a wide range of metrics (SNR, SI-SDR, STOI, and PESQ) and SNR levels between -30 dB and 0 dB. These gains were observed across both time-domain and time-frequency domain models, confirming that rotor-informed conditioning is broadly applicable regardless of architectural family or representation domain. A lightweight rotor-conditioned variant of DCCRN achieved best or near-best performance in several settings despite using only 25% of the parameters, demonstrating that rotor integration can yield high efficiency without sacrificing quality. These findings establish rotor-informed conditioning as an effective and generalizable strategy for robust speech enhancement in UAV contexts. Overall, the approach shows strong potential for embedded, real-time drone audition systems. Future work will focus on adapting these models to causal and low-latency conditions, quantifying computational demands for onboard inference, and validating performance on real-world drone recordings beyond synthetic mixtures.

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Authors' contributions

AG conceptualized, developed, prototyped, tested, and evaluated the algorithm and performed the statistical analysis. AG also prepared the manuscript. FF conceptualized the method, reviewed and edited the manuscript, and coordinated the research project. CD developed and fine-tuned the algorithm. AG and CD performed the audio recording and RPS transcriptions. GF curated and realized the experimental setup. DS proposed the algorithm evaluation metrics, suggested the quantitative and perceptual measures, and identified relevant literature references. FF, DS, and CD reviewed the manuscript. All authors read and approved the final manuscript.

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Data availability

The source code, the recordings made at the ALVAR lab, and some audio examples can be downloaded at: <https://github.com/gullogullo/Rotors-informed-Wave-U-Net-for-Speech-Enhancement>. The dataset generated during the current study is built from the publicly available DRone EGonoise and localization dataset (DREGON) (<http://dregon.inria.fr/datasets/dregon/>) and the Diverse Environments Multichannel Acoustic Noise Database (DEMAND) (https://dcase-repo.github.io/dcase_datalist/datasets/scenes/demand.html).

Declarations

Competing interests

The authors declare that they have no competing interests.

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